

Question	Marking Guidance	Mark	Comments
03.1	1. Heat and (then) weigh; 2. (Until) mass is constant;	2 (2 x AO3)	Accept 'heat until sample remains the same mass' for 2 marks 1. Accept any specified temperature for 'heat' 1. Ignore length of time heated 1. Reject 'burn' 2. Accept weight for mass

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03.2	7.6 / 7.61 / 7.608 / 7.6076 (kJ g⁻¹) = 2 marks;; Incorrect, but answer shows 76 (ignore any subsequent numbers, position of decimal point and preceding/subsequent zeros i.e. incorrect conversion) = 1 mark OR Incorrect answer of 11.4 (kJ g⁻¹) (ignore any subsequent numbers i.e. no division by 1.5.) = 1 mark OR Incorrect answer of 0.05 (kJ g⁻¹) (ignore any subsequent numbers i.e. not multiplied by 150) = 1 mark OR Incorrect answer of 1.8 (kJ g⁻¹) (ignore any subsequent numbers i.e. not multiplied by 4.18) = 1 mark OR Incorrect answer of 0.418 / 0.42 (kJ g⁻¹) (ignore any subsequent numbers i.e. not multiplied by 18.2) = 1 mark OR Incorrect answer of 17 (kJ g⁻¹) (ignore any subsequent numbers i.e. multiplied rather than divided by 1.5) = 1 mark;	2 (2 x AO2)	Accept 8 for 2 marks .

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03.3	1. Net primary production = gross primary production minus respiratory losses OR $NPP = GPP - R$; 2. (At higher temperature) photosynthesis/GPP decreases OR (At higher temperature) respiration increases OR (At higher temperature) enzymes for photosynthesis denature;	2 (1 x AO1, 1 x AO2)	1. Accept 'is' for = sign 1. Accept RL (respiratory losses) for R 1. Accept respiration for 'respiratory losses' 2. Accept 'respiratory losses' for respiration 2. Reject enzymes for photosynthesis and respiration denature 2. Reject 'more energy used in respiration'